



Clinical heterogeneity of juvenile myoclonic epilepsy: Follow-up after an interval of more than 20 years



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ABSTRACT

Purpose: The view that juvenile myoclonic epilepsy (JME) is a uniform and life-long disorder is currently being challenged. The aim of this study was to assess the seizure and psychosocial outcome of JME at least 20 years after onset.

Methods: In 1992, 42 patients with JME were identified. In 2012, 37 agreed to a semi-structured interview. In the remaining five, only medical records were available.

Results: Of 40 patients with known seizure outcome, 21 were in remission for >5 years. Seven were off antiepileptic drugs (AEDs), four being seizure free for >10 years. Myoclonic seizures (MC) evolving to generalized tonic–clonic seizures (GTC) were associated with seizure persistence ($p = 0.013$), whereas >1 year between MC and GTC onset was associated with a trend to GTC remission ($p = 0.069$). Of 19 patients with uncontrolled seizures, eight experienced remission with second generation AEDs.

Favorable psychosocial outcome by interview was found in a third, whereas another third had psychiatric comorbidity, seven with substance or alcohol abuse. Psychosocial and seizure outcome did not correlate.

Conclusion: This study corroborates the heterogeneity of JME in terms of seizure and psychosocial outcome, but without a clear association between the two. It confirms that seizure control may persist after AED withdrawal in some and supports MC evolving to GTC as a predictor of seizure persistence. Moreover, it suggests that newer broad spectrum AEDs may improve the prognosis of JME; their impact should be focus of prospective studies.

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1. Introduction

Juvenile myoclonic epilepsy (JME) is the most common type of genetic (idiopathic) generalized epilepsy, comprising 5–10% of all epilepsies.¹ Onset is usually in the second decade, and the cardinal symptom is early-morning myoclonic seizures (MC), often precipitated by sleep deprivation. Approximately 90% of patients have generalized tonic–clonic seizures (GTC), and one third has absence seizures.²

Although response to appropriate treatment is good for most patients,^{3,4} JME has been considered a lifelong disorder with a high risk of seizure relapse on discontinuation of antiepileptic drugs (AEDs).^{5–7} Nevertheless, four long-term follow-up studies have

shown that some patients may stop AED treatment and remain seizure free.^{8–11}

In contrast to the seemingly positive seizure outcome in the majority, there is expanding evidence of an unfavorable psychosocial outcome in many patients.^{9,12} This has been ascribed to a subtle frontal lobe dysfunction in JME.^{13,14}

As long-term studies of JME are scarce, there is a need for more clinical research on the course and prognosis of this syndrome. Thus, we aimed to assess the severity of the disorder in a cohort of patients after at least 20 years of follow-up, and to study clinical characteristics in relation to seizure and psychosocial outcome.

2. Methods

In 1992, consecutive patients with JME at Trondheim University Hospital were identified for recruitment in a study on the association with human leukocyte antigens.¹⁵ The clinical diagnosis of JME was based on the 1989 classification of epilepsies and epileptic syndromes by the International League Against

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Table 1

Number of patients included in the various analyses after follow-up >20 years.

	Original cohort (1992)	Seizure outcome	Psychosocial outcome	Disability benefits
Interviewed	37	37	37	37
Unavailable	2	1	0	1
Deceased	3	2	0	2
Total	42	40	37	40

Epilepsy (ILAE).¹⁶ In 2012, we attempted to contact all 42 patients with a request for participation in a follow-up study.

One patient could not be included as he died after only 13 years of follow-up. Two other deceased patients followed for more than 20 years were included based on their medical records. Two patients could not be contacted as they were currently registered with no fixed address; in one medical records were sufficient for inclusion (Table 1).

37 patients (88%) were available for a standardized semi-structured telephone interview, focusing on seizure types, seizure frequency, medication and psychosocial outcome. Medical records were reviewed for background information.

Psychosocial outcome was assessed based on five elements: graduation from high school, disability or unemployment benefits, alcohol or illegal substance abuse causing social problems, need of psychiatric health care and self-reported social network. Social network, as well as perceived satisfaction with health, was simply graded as poor, sufficient or good, as more complex scales were considered less useful for telephone interviews.

Chi-square test and independent *t*-test were used for comparison of categorical and continuous variables, respectively. $p < 0.05$ was considered significant.

The study was approved by the Regional Committee for Medical Research Ethics, and written informed consent was provided by all interviewed participants.

3. Results

3.1. Patients

Of the 42 patients forming the basis of this study, 25 (60%) were women. Forty patients (95%) had GTC and 12 (29%) had absences. One patient had MC only. In five patients, GTC was the first reported seizure type. The duration of JME, calculated from onset of the first motor seizure (MC or GTC) was 31 ± 9.3 years. Mean age at JME onset was 16 ± 3.2 years (women 15 ± 3.1 , men 17 ± 3.0 , $p = 0.056$). In three female patients, absences were the first seizure type, with onset at age eight, nine and 12 years, respectively, whereas MC started at age 18 in the two first, and at age 19 in the third.

In the three deceased patients, causes of death were suicide (age 28), cancer (age 59) and dementia (age 81).

3.2. Seizure outcome

In 40 patients seizure outcome was known at least 20 years after JME onset. Median age at follow up was 47 (range 35–81) years.

The five and ten-year seizure remission rates for GTC and MC are shown in Table 2. More than 50% had been completely seizure-free for the last five years, and one-third for at least ten years. Altogether 19 patients had persistent seizures within the last five years; one had GTC only, ten had MC only and eight had both seizure types. Of the 18 interviewed patients with persistent MC, all but two (age 39 and 41) reported a reduced frequency compared to younger years.

Table 2

Terminal seizure remission in 40 patients with JME after follow-up > 20 years.

Seizure remission	Five years	Ten years
Generalized tonic–clonic seizures	31 (78%)	24 (60%)
Myoclonic seizures	22 (55%)	16 (40%)
Both seizure types	21 (53%)	13 (33%)

Table 3

Clinical characteristics in relation to seizure outcome in 40 patients with JME followed >20 years.

	5-year seizure remission	Seizures within last 5 years	P-value
Three seizure types	5 (50%)	5 (50%)	0.855
Absence seizures	5 (45%)	6 (55%)	0.583
Myoclonic seizures not strictly confined to awakening	6 (46%)	7 (54%)	0.564
Myoclonic seizures preceding GTC	8 (35%)	15 (65%)	0.013
>1 year between onset of myoclonic seizures and GTC	9 (60%)	6 (40%)	0.464

None of the three patients with absence epilepsy prior to MC onset had experienced seizure remission. All were females. Two had persistent MC and GTC; one had been without GTC for more than ten years, but still had occasional MC without taking AEDs.

The relationship between various clinical characteristics and five-year seizure remission is shown in Table 3. MC habitually or occasionally preceding GTC (clonic–tonic–clonic seizure) was associated with a poorer seizure outcome. Fifteen patients had a delay of more than one year between MC onset and first GTC. At follow-up, 11 of them had not had GTC for more than ten years ($p = 0.069$). However, the longer interval between MC and GTC onsets, did not influence five year complete seizure remission (Table 3). There was no difference in age at JME onset between patients with five-year seizure remission and patients with ongoing seizures (16 ± 3.9 years vs. 16 ± 2.3 years, $p = 0.494$). Eleven women (44%) and ten men (67%) had been seizure-free for more than five years ($p = 0.165$). Seizure persistence was not associated with poor satisfaction with health ($p = 0.506$).

3.3. Psychosocial outcome

Fourteen of the 37 interviewed patients (38%) had a favorable psychosocial outcome, meaning they had graduated from high school, were working, had no history of a psychiatric disorder, had a good or sufficient social network and no substance or alcohol abuse. Altogether 26 (70%) had graduated from high school, and 12 (32%) were college or university graduates, one with a PhD degree. There was no association between these factors and seizure outcome (Table 4).

We were able to obtain information on employment and disability benefits from 40 patients. Twenty-three patients (58%)

Table 4

Unfavorable psychosocial factors in relation to seizure outcome in 37 patients with JME after >20 years.

	5-year seizure remission	Seizures within last 5 years	P-value
No high school graduation	4 (36%)	7 (64%)	0.087
Disability benefits/unemployment	7 (50%)	7 (50%)	0.898
Substance abuse	3 (75%)	1 (25%)	0.316
Ever needed psychiatric health care	7 (54%)	6 (46%)	0.823
Poor self-reported social network	2 (40%)	3 (60%)	0.162
No unfavorable psychosocial factors	7 (50%)	7 (50%)	0.699

Table 5

The use of second generation antiepileptic drugs in 40 patients with JME followed >20 years.

AED	Ever used	Current use	Retention rate (%) ^a
Levetiracetam	17	10	59
Lamotrigine	16	8	50
Topiramate	11	6	55

^a The rate of patients with continued use of the drug for >1 year at follow-up.

were employed, including two self-employed, and three were unemployed. Fourteen (35%) received disability benefits, but only half of them due to epilepsy alone; other causes included psychiatric disorders, chronic pain, multiple sclerosis and rheumatoid arthritis.

Thirteen of the interviewed patients (35%) had received psychiatric care, nine (24%) for depression, four for anxiety and two for psychosis. Four admitted alcohol or illegal substance abuse. According to medical records, one of the patients with no fixed address and two of the deceased also had such problems. Four of these patients had been convicted prisoners.

At interview, 32 patients (86%) reported their social network to be good (18 patients) or sufficient (14 patients). As many as 28 (76%) were satisfied (23 patients) or very satisfied (five patients) with their health, including three of the seven receiving disability benefits mainly for comorbid conditions. Of the nine patients who were unsatisfied, four had a good seizure outcome. Even in the 18 interviewed patients with occurrence of seizures during the last five years, the majority were satisfied (ten patients) or very satisfied (three patients) with their health.

There was no association between psychosocial outcome and gender, age at onset or duration of epilepsy prior to initiation of treatment (data not shown).

3.4. Treatment and follow-up

Seven patients (18%) had stopped AED treatment. Four of them had been completely seizure-free for more than ten years after withdrawal. Two still had occasional MC, but no GTC. One had a GTC within the last year, but reported several years between seizures. Four of the seven patients who had discontinued medication, did so on their own initiative, two without medical supervision.

Of those remaining on treatment, 17 patients used monotherapy, and 18 a combination of AEDs. Among the 17 patients with five-year seizure remission still using AEDs, 10 had monotherapy, mainly valproate (seven).

Valproate had been used by all, apart from three females. The second generation broad-spectrum AEDs, lamotrigine, levetiracetam and topiramate had been tried in a total of 21. The retention rates, denoting continued use >1 year at follow-up, are shown in Table 5. Of the 19 patients who had received second generation AEDs for uncontrolled seizures for more than five years, eight obtained seizure remission, four of them after the introduction of levetiracetam. The remaining patients used lamotrigine in

combination treatment. However, lamotrigine was withdrawn in five other patients due to more frequent MC. Levetiracetam, in contrast, was specifically reported to reduce MC in four, but treatment was stopped in another five due to an apparent GTC increase. Topiramate was considered beneficial in the six patients remaining on treatment, partly by counteracting valproate-induced weight gain.

Thirteen patients (33%) had not seen a neurologist for more than ten years. Only seven of them had been seizure-free for more than five years. Two had a GTC less than one year earlier. One had several unfavorable social factors, including substance abuse. Beyond that, we found no association between drop-out from follow-up and social outcome.

4. Discussion

4.1. Seizure outcome

This study contributes to the knowledge on the long-term seizure prognosis of JME by investigating 40 patients after a mean of 31 years after onset. After at least 20 years, more than half of the patients had been seizure-free for at least five years. Seven had stopped using AEDs, and four had remained seizure-free off medication for as long as ten years, suggesting epilepsy as no longer present.

To our knowledge, four other studies have followed JME cohorts for decades, all of them published since 2008^{8–11} (Table 6). Due to different methodology, these studies are only partly comparable. A limited number of patients were included in each study, and different aspects of the disorder were addressed.

The most extensive long-term study, with regard to both patient number and follow-up time, is from Berlin, reporting on patients originally selected by Dieter Janz, the first to describe JME as a specific syndrome. Five-year seizure remission was just below 60%, and nearly one fifth remained seizure-free off medication.¹¹ This is in accordance with another recent study from Germany.¹⁰ However, in the latter study, more than half of the screened patients were lost to follow-up, presumably leaving out a number of patients with poor prognosis. A strength of our study is the very low drop-out rate.

Baykan et al. reported 48 patients from Turkey attending regular visits to an epilepsy center for an average of 20 years. According to their findings, MC subside in the fourth decade of life. More than 50% no longer had MC at follow-up.⁸ Our findings were similar; all but two of our patients had experienced a considerable improvement of MC, and more than half were in complete remission. MC may sometimes be under-reported, as not all patients interpret single jerks as seizures. Like Baykan et al., we specifically asked for MC after describing such seizures carefully.

Five-year seizure remission in the Turkish study was low (27%). An inclusion criterion was at least seven years of follow-up at the epilepsy clinic prior to inclusion. Thus, the selection of participants appeared to be biased toward patients with more severe epilepsy actively seeking medical advice and follow-up. Our study is also hospital-based and may accordingly be biased toward difficult-to-treat epilepsy as well, but it also includes patients dropping out of

Table 6

Comparison of long-term follow-up studies of JME.

	Design	Patient number	Lost to follow-up	Mean follow-up (years)	Mean age at onset	5-year seizure remission	Patients off medication	5-year seizure remission off medication
Baykan et al. (2008)	Hospital-based	48	Not reported	20	14	13 (27%) ^a	9 (19%)	4 (8%)
Camfield and Camfield (2009)	Population-based	23	4%	26	10	7 (30%)	11 (48%)	6 (26%)
Geithner et al. (2012)	Hospital-based	31	56%	39	13	21 (68%)	6 (19%)	6 (19%)
Senf et al. (2013)	Hospital-based	66	19%	45	14	39 (59%)	12 (18%)	11 (17%)
Present study	Hospital-based	40	5%	31	16	21 (53%)	7 (18%)	4 (10%)

^a At any time during follow-up; the other figures denote terminal remission.

regular specialist care. Conceivably, more patients with seizure control are covered, as well as those with poor psychosocial outcome. Nevertheless, common findings in these long-term hospital-based studies are the lower seizure remission rates compared to studies on new onset or valproate-naïve patients (80%).^{4,17} It is noteworthy that studies with the longest follow-ups show the best seizure outcomes (Table 6), a finding which may reflect a general improvement with age.

The only population-based study⁹ differed from the other studies in several respects, including eager attempts to discontinue treatment and inclusion of only patients with pediatric seizure onset age (mean ten years). As many as 18 of the 23 patients tried AED withdrawal after initial successful treatment. Eleven remained off medication, but only six stayed completely seizure free; five were left with rare or non-intrusive seizures. Only one patient among those who never tried to stop treatment achieved terminal remission exceeding 5 years (Table 6).

Moreover, patients with prior absence epilepsy progressing to JME were excluded from that study, as this subgroup may have a different outcome. Female preponderance and poor treatment response have been pointed out as characteristics of childhood absence epilepsy progressing to JME.⁷ In our study, all three patients with absence epilepsy evolving to JME were females with persistent seizures, contributing to a gender skewed seizure outcome. In the Berlin study, additional absence seizures with onset at 12 years or later were also a negative predictor of seizure outcome.¹¹ Differing results in various studies may partly be due to the limited number of patients in each of them; in addition, the subtle manifestations of absences may sometimes have been neglected.

The majority of patients not taking AEDs in the present study had discontinued treatment on their own initiative. Others report similar findings.¹⁰ Results from the five long-term studies of JME published to date, including our own, suggest that the current common practice of keeping all patients on life-long AED treatment should be less strict (Table 6).

We found a trend to an association between GTC remission and a longer interval between onset of MC and the first GTC. It is not surprising that a delay of the occurrence of GTC denotes a lower tendency to later GTC. If supported by further studies, it might nevertheless serve as a useful predictor of a subsequent remission of the most intrusive seizure type. Like Geithner et al., we found that MC evolving to GTC predicted an unfavorable seizure outcome.¹⁰

No previous studies have addressed the effect of the new generation of broad spectrum AEDs on the long term outcome of JME. An appropriate assessment of sustained drug efficacy was not possible in this study, partly as background medications were often changed, and probably also due to the development of the disorder itself and the lessened impact of seizure-provoking factors with age. Nevertheless, half of the patients in the present study had tried second generation AEDs for uncontrolled seizures, and 40% of them subsequently achieved at least five-year seizure remission. The highest retention rate at final follow up was found for levetiracetam (Table 5), which is suggested to have a specific antimyoclonic effect.³ After introduction of levetiracetam, seizure freedom was obtained in 20–30% of patients, similar to findings in a previous prospective study.¹⁸ However, this drug should be used with care, as some patients experience an increased tendency to GTC. Remarkably, levetiracetam and lamotrigine may have an opposite effect on MC.³

4.2. Psychosocial outcome

Several studies have confirmed that many people with JME have a peculiar personality profile and a trend toward social maladjustment,^{14,19,20} but few long-term studies have focused on the relation of seizure and psychosocial outcome. One third of the

present patients had at some time received psychiatric health care, but no correlation with long-term seizure outcome was found (Table 4). Previous studies have reported an increased risk of psychiatric comorbidity in JME.^{19,21} A French group identified a clear association between the presence of psychiatric disorders and AED-resistant seizures in newly diagnosed patients²², a finding which may have been related to ongoing psychiatric problems and an early stage of the disorder. This finding has been supported by other studies.²¹ The different results in our study may be due to the natural course of JME and its treatment. Moreover, the current development of social welfare facilitating social and vocational integration in our part of the world may have contributed to a reduced interaction between seizures and a history of psychiatric disorders.

Social problems, impulsivity, sleep/wake rhythm, decision-making and risk-taking behavior in JME have been highlighted by several authors.^{12,14} Dieter Janz, in his early reports, considered these patients to have "...an engaging, but emotionally unstable, fairly immature personality..."⁵ Some patients with JME have a tendency to non-compliance concerning both drug adherence and seizure provoking factors, as well as to missing out on outpatient appointments.^{8,20} Nevertheless, some aspects of these characteristics may be perceived as attractive.²³ Conceivably, this personality profile might explain why most patients with uncontrolled seizures reported satisfaction with health.

A predominance of these typical personality traits in patients with ongoing seizures within the last year has recently been shown,¹⁴ but was not reflected as an association between seizure and psychosocial outcome in the present study (Table 4). These features have received ample attention in recent literature. The main focus of interest has been a JME-associated frontal lobe impairment, demonstrated both on neuropsychological tests and by sophisticated imaging techniques.¹³ Moreover, the juvenile onset of the disorder affects people in a vulnerable phase of life concerning education, identity-formation and socializing. In accordance with findings in older patients with JME,²⁴ it is our clinical impression that these problems prevail in the early stage of the disorder. The development of the personality profile with age should be the focus of future prospective long-term studies.

Camfield and Camfield also emphasized that several of their patients had an unfavorable psychosocial outcome, including depression and unemployment.⁹ A history of substance or alcohol abuse was present in many patients (17%) in our original cohort, and in some even a criminal record (10%). One of these patients had died from suicide. The use of illegal drugs and alcohol in patients with JME should receive attention, both clinically and in future research. Moreover, specialist follow-up should not exclusively be left to the patients' own initiative, as they may exhibit a striking indifference to the consequences of the disorder. Seven of our patients had not seen a neurologist for more than ten years, in spite of ongoing seizures.

The lack of a comparable non-JME control group and a standardized questionnaire for collection of psychosocial data are clear limitations of this study. The fact that few subjects of the original cohort were lost to follow-up is a strength; however, small hospital-based studies cannot truly reflect all aspects of the average JME population.

Nevertheless, this study demonstrates that a substantial number of people with JME seem to manage well. Negative psychosocial elements were absent in more than one third of our patients. Many were financially independent with a satisfactory educational achievement; one was even a PhD graduate. In addition to the great diversity of social ability, and the well-known differences in response to treatment, neuropsychological testing has also demonstrated variable and slightly contradictory results.¹³ The heterogeneity of JME is also corroborated by genetic

findings; five different Mendelian genes and several susceptibility loci have currently been identified.²⁵

5. Conclusion

The clinical diversity of JME is remarkable, but so far we are left with few significant early prognostic predictors. The present study confirms that MC evolving to GTC is associated with a poor seizure outcome. It suggests that a long period with only MC prior to the first GTC, may indicate a future remission of GTC. The new broad spectrum AEDs appear to have the potential to reduce the proportion of pharmacoresistant patients; levetiracetam is particularly promising. Moreover, current findings support a more relaxed attitude to the recommended life-long treatment of JME, as a number of patients remain in remission after AED withdrawal.

Many people with JME are burdened by psychosocial maladjustment, including psychiatric comorbidity, in some cases even substance and alcohol abuse, as well as criminal behavior. Efforts to counteract this dangerous development in young people at risk should be part of an active and supportive comprehensive epilepsy service.

Notwithstanding, we wish to emphasize that a large number of patients in this study had a favorable psychosocial outcome, often independent of seizure outcome. A stigma of negative expectations should not be transferred to these patients, as they appear to belong to a variant of JME devoid of a manifest adverse personality profile.

Conflict of interest statement

Marte R. Syvertsen has received honorarium from GlaxoSmithKline. Eylert Brodtkorb has received honoraria and financial support for attending conferences from GlaxoSmithKline, UCB and Eisai. Bendicte S. Stordrange and Selma Thuve declare no conflict of interest.

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